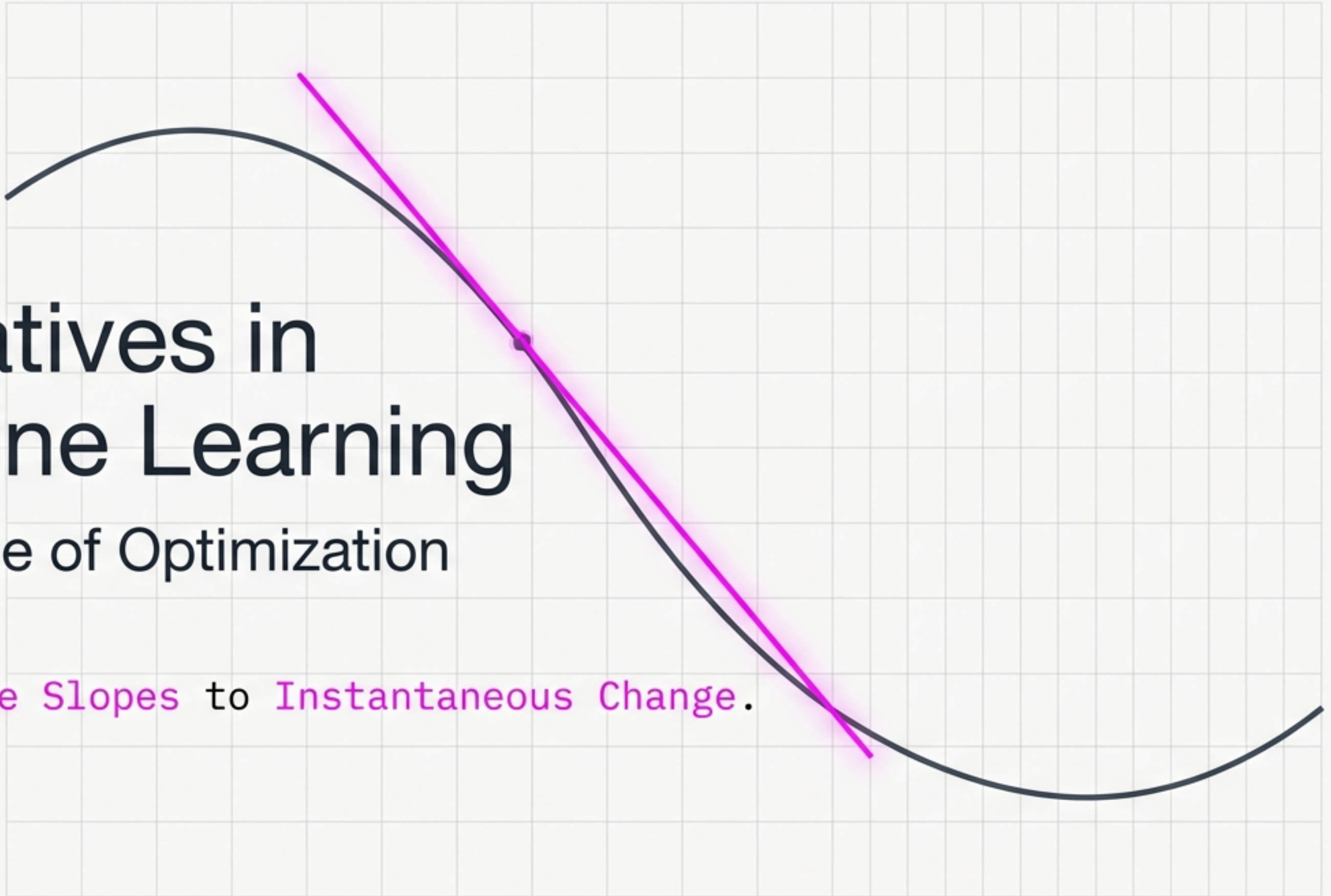


Derivatives in Machine Learning

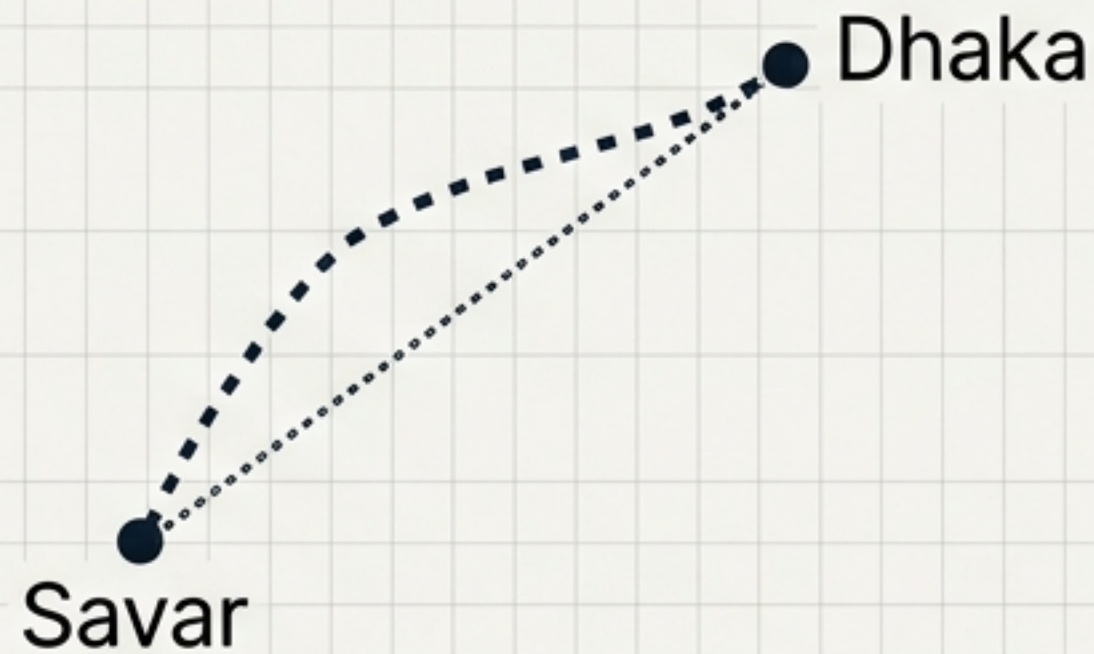
The Engine of Optimization

From Average Slopes to Instantaneous Change.



What is a Derivative?

A derivative calculates the exact rate of change at a specific, single moment.



Average Speed: 40 km/h



10:05 AM

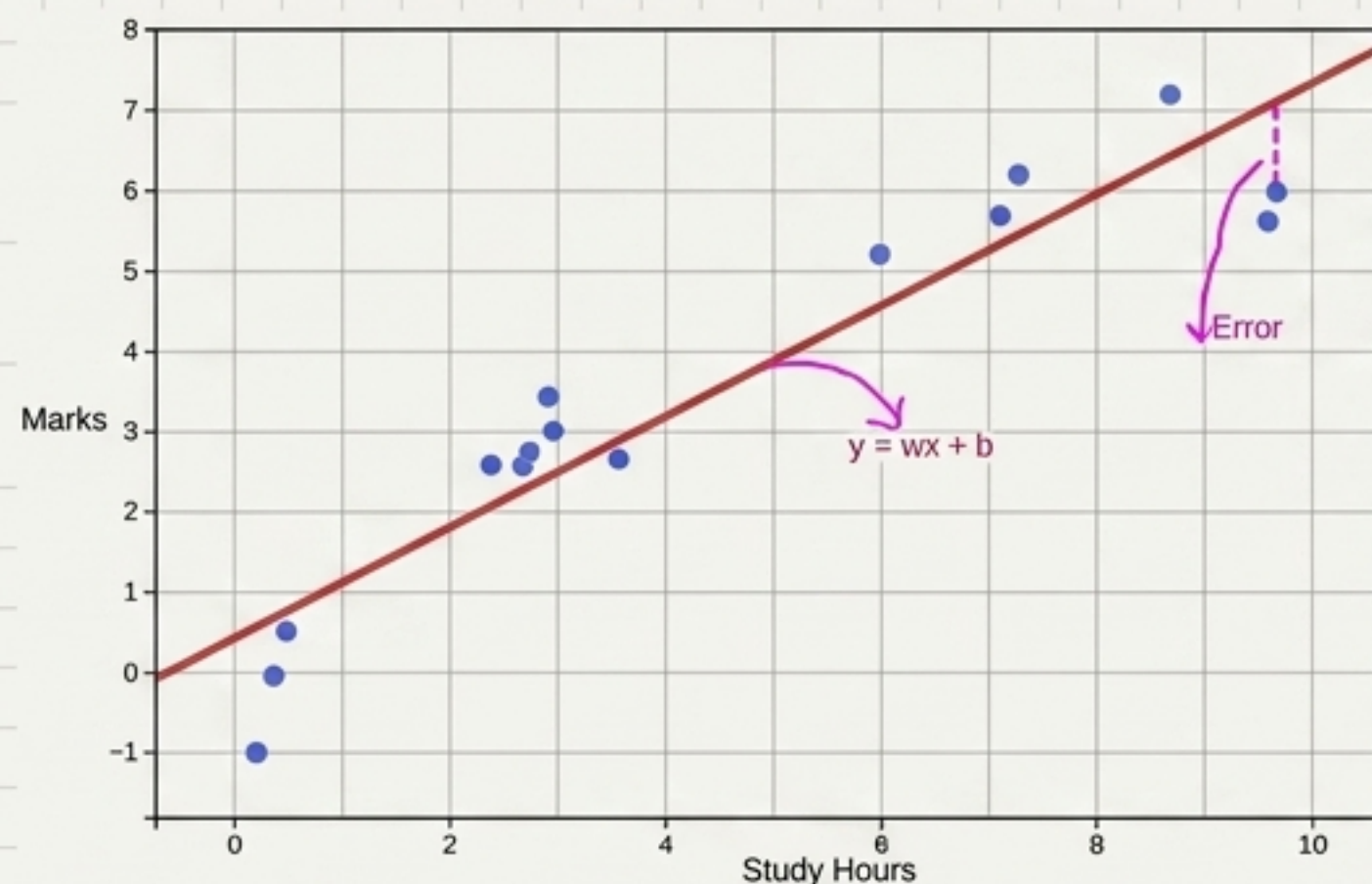
Instantaneous Speed: 60 km/h

The ML Translation: If I change a specific variable (like **Weight w**) right now by a tiny fraction, exactly how much will my output change?

Measuring Mistakes: Loss vs. Cost

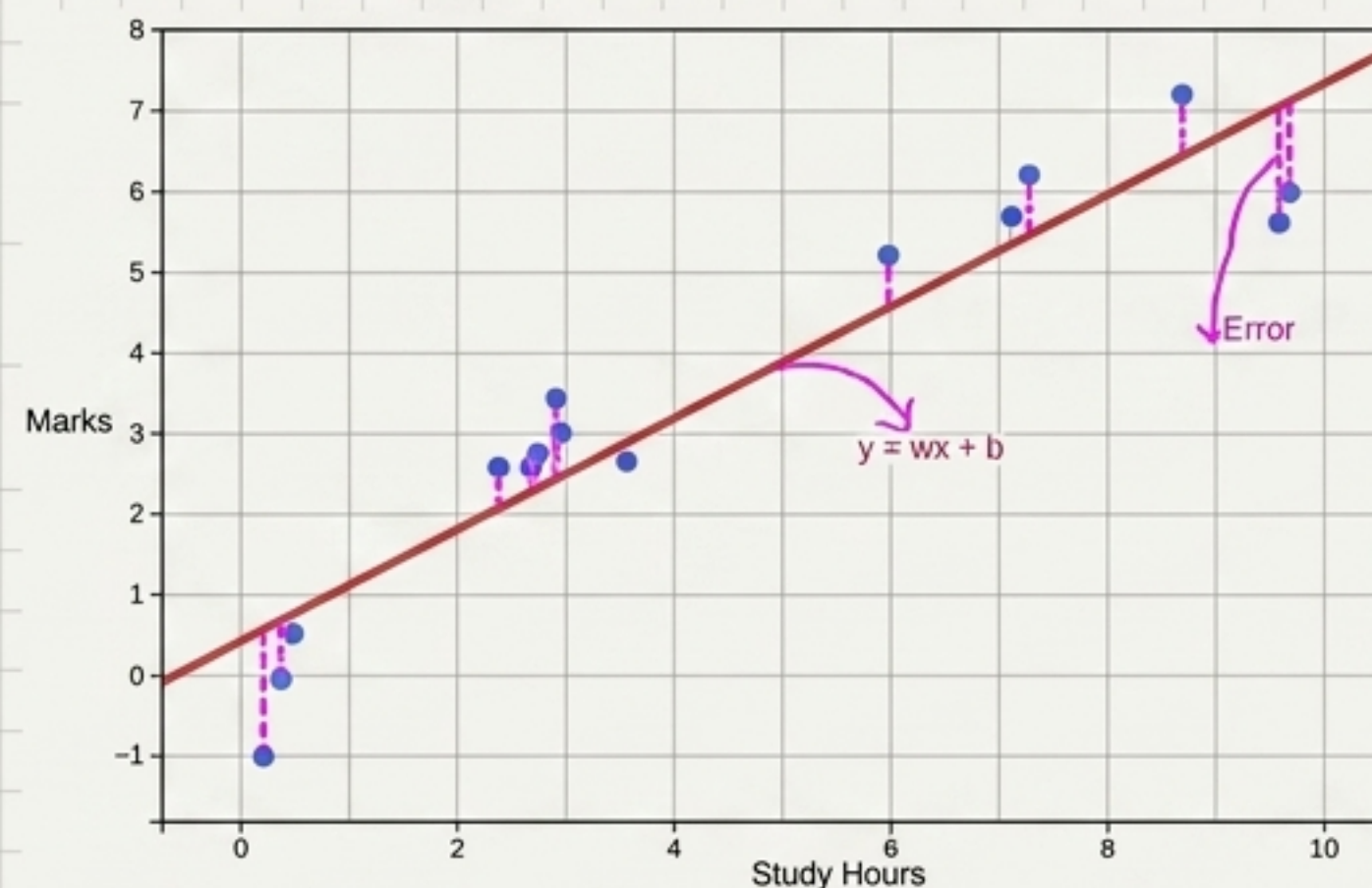
Red Line = Prediction ($y = wx + b$). Blue Dots = Real Data.

Loss Function



The error calculated for a single, individual data point.

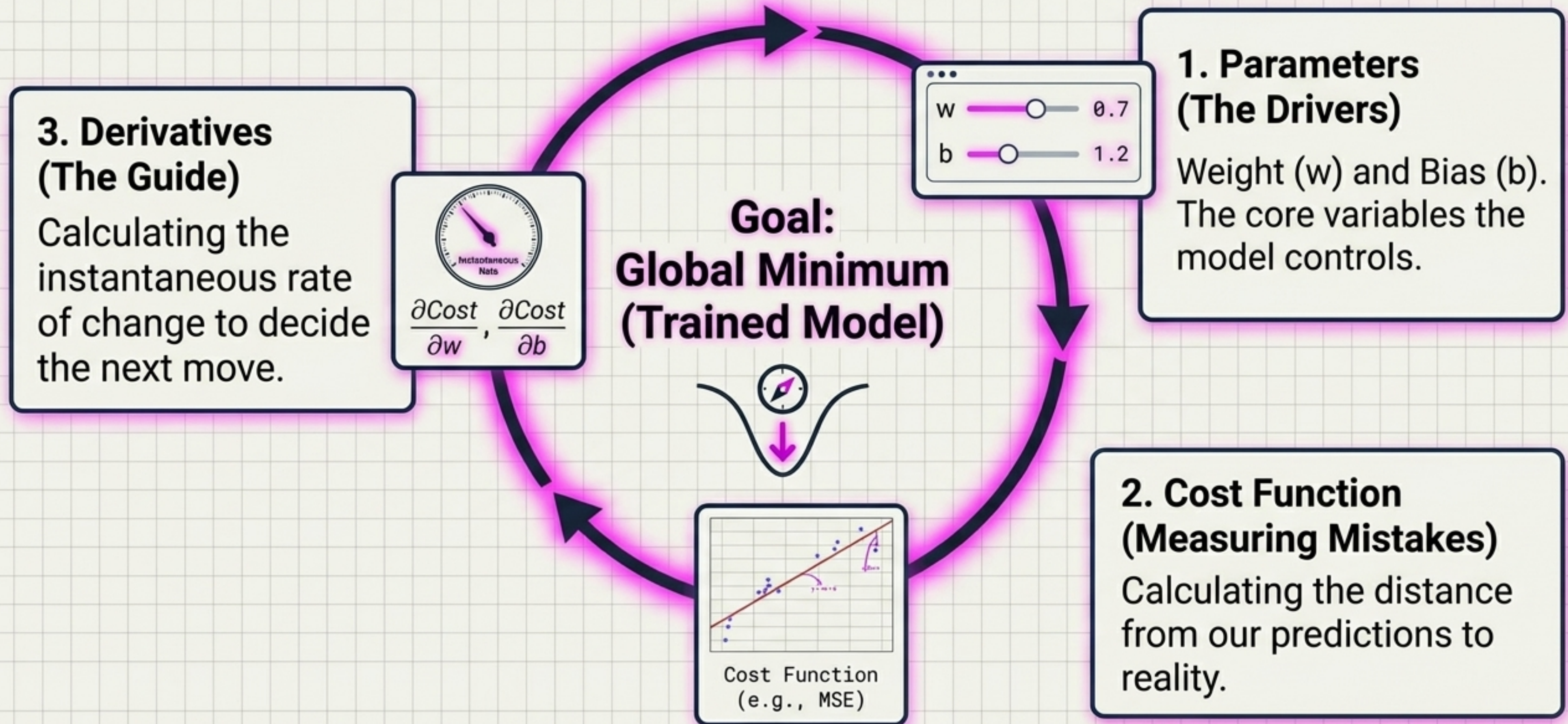
Cost Function



The total or average error across the entire dataset.

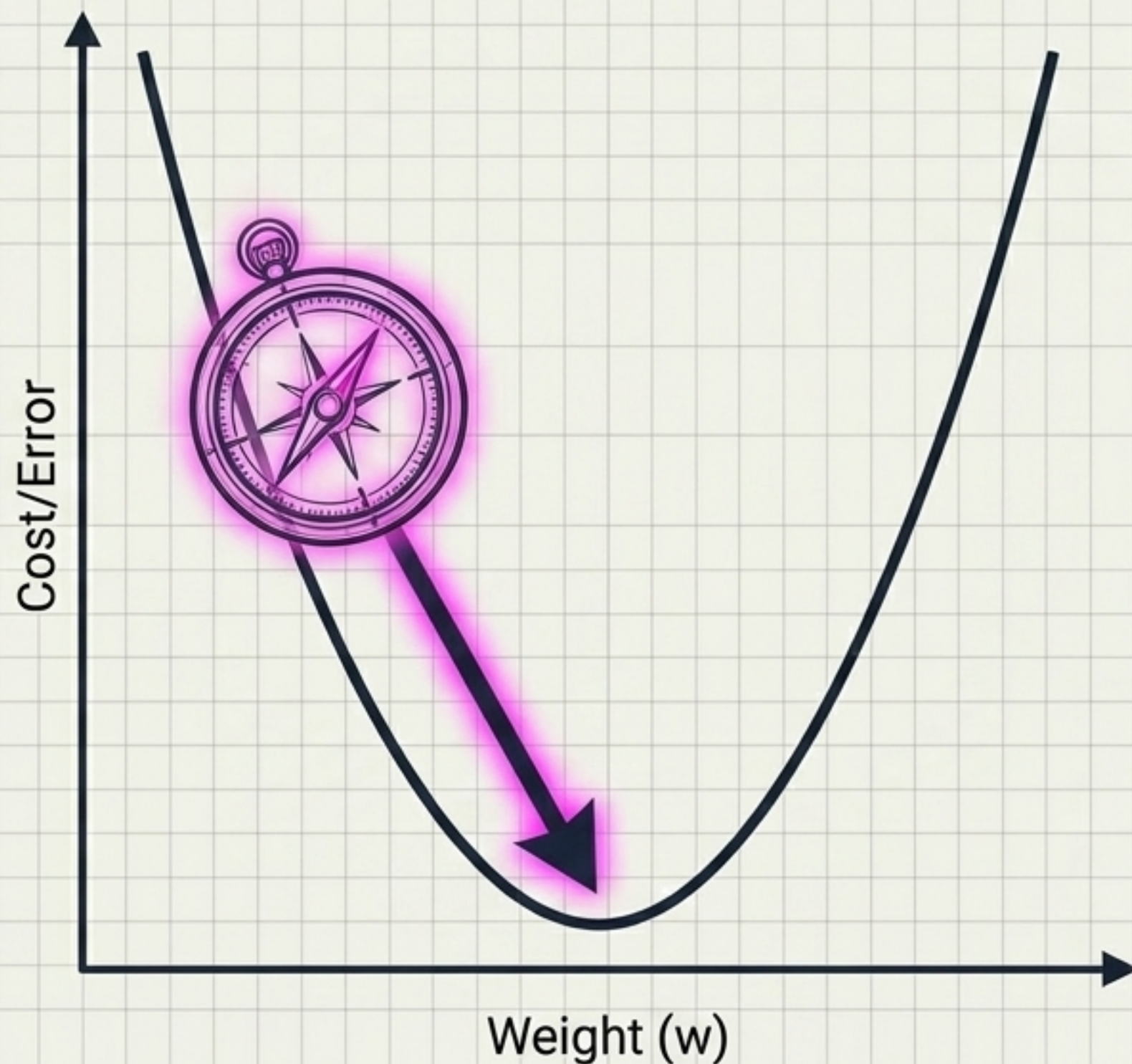
Optimization Goal: Shrink the dashed lines. Minimize the Cost Function.

The 3-Step Learning Loop



The Optimization Compass

The **derivative** calculates the **steepness** of the error curve at any exact moment.

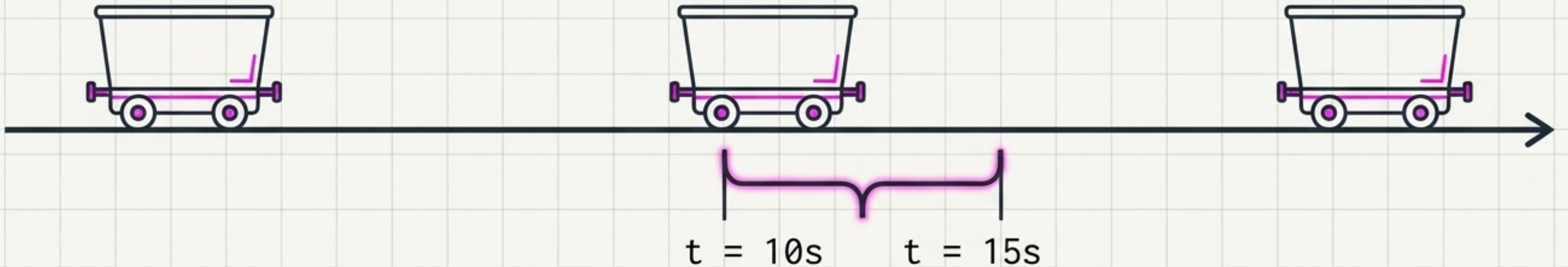


The Optimizer's Logic

“If you increase the **weight**, the **cost** will go up. So, **decrease** the **weight** to reach the **minimum error**!”

The Birth of Derivatives: Shrinking the Interval

A car's position is recorded over time.



$$\frac{\Delta d}{\Delta t} = \frac{(28 - 18)}{(15 - 10)} = \frac{10}{5} = 2 \text{ m/s}$$

The Problem: Average velocity hides the truth. It assumes the car moved at exactly **2 m/s** the entire time, ignoring **speeding up** or slowing down.

Approaching Zero: From Δ to d

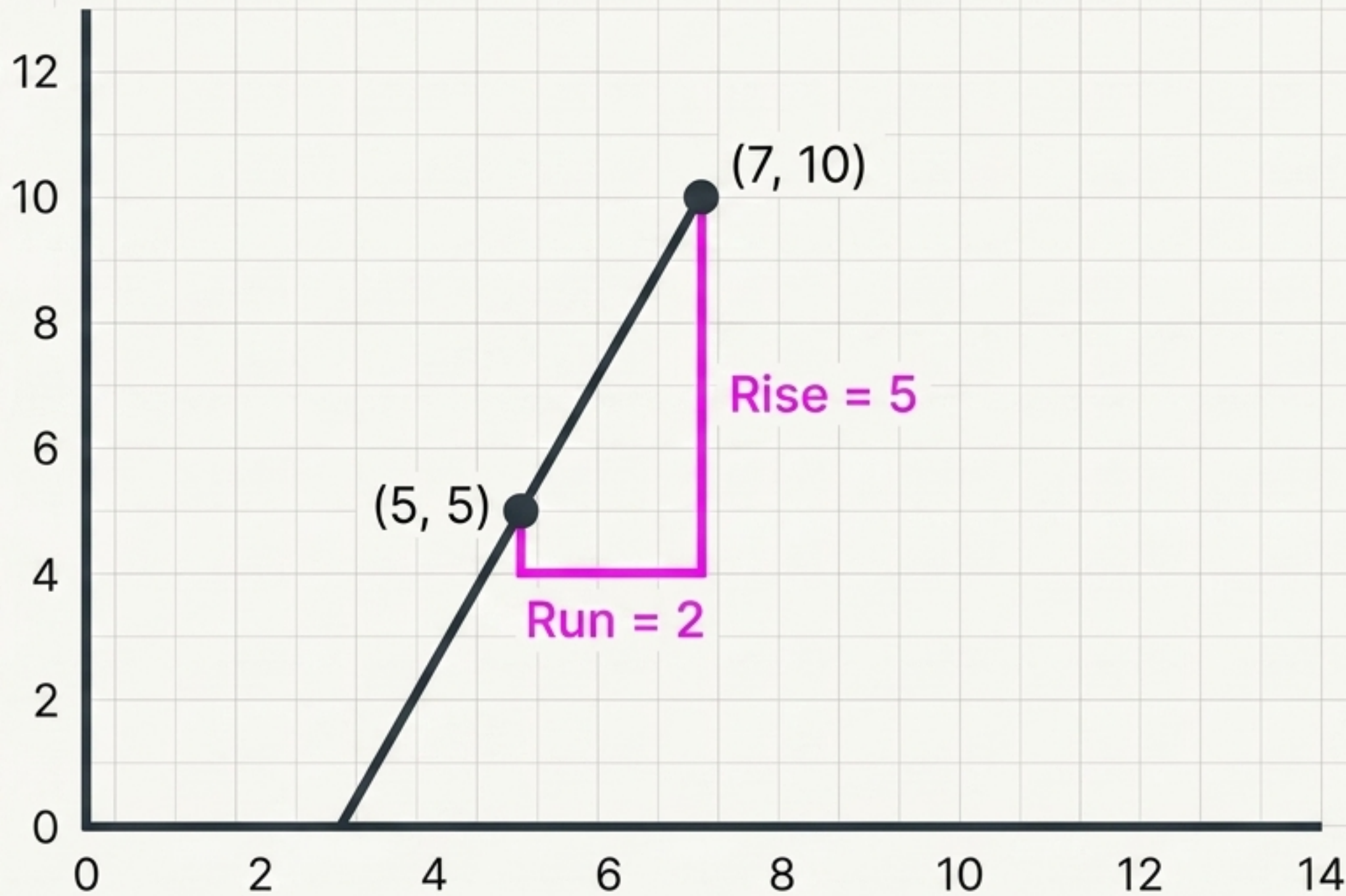
Interval: 5 seconds	Velocity: Average	Symbol: $\Delta d / \Delta t$
Interval: 1 second	Velocity: Closer to reality	Symbol: $\Delta d / \Delta t$
Interval: 0.001 seconds (Infinitesimal)	Velocity: Instantaneous	Symbol: dd / dt

Translation Matrix

Physics	Machine Learning
Δ Distance Δ Time $\rightarrow dd / dt$	Δ Error Δ Weight $\rightarrow \partial E / \partial w$

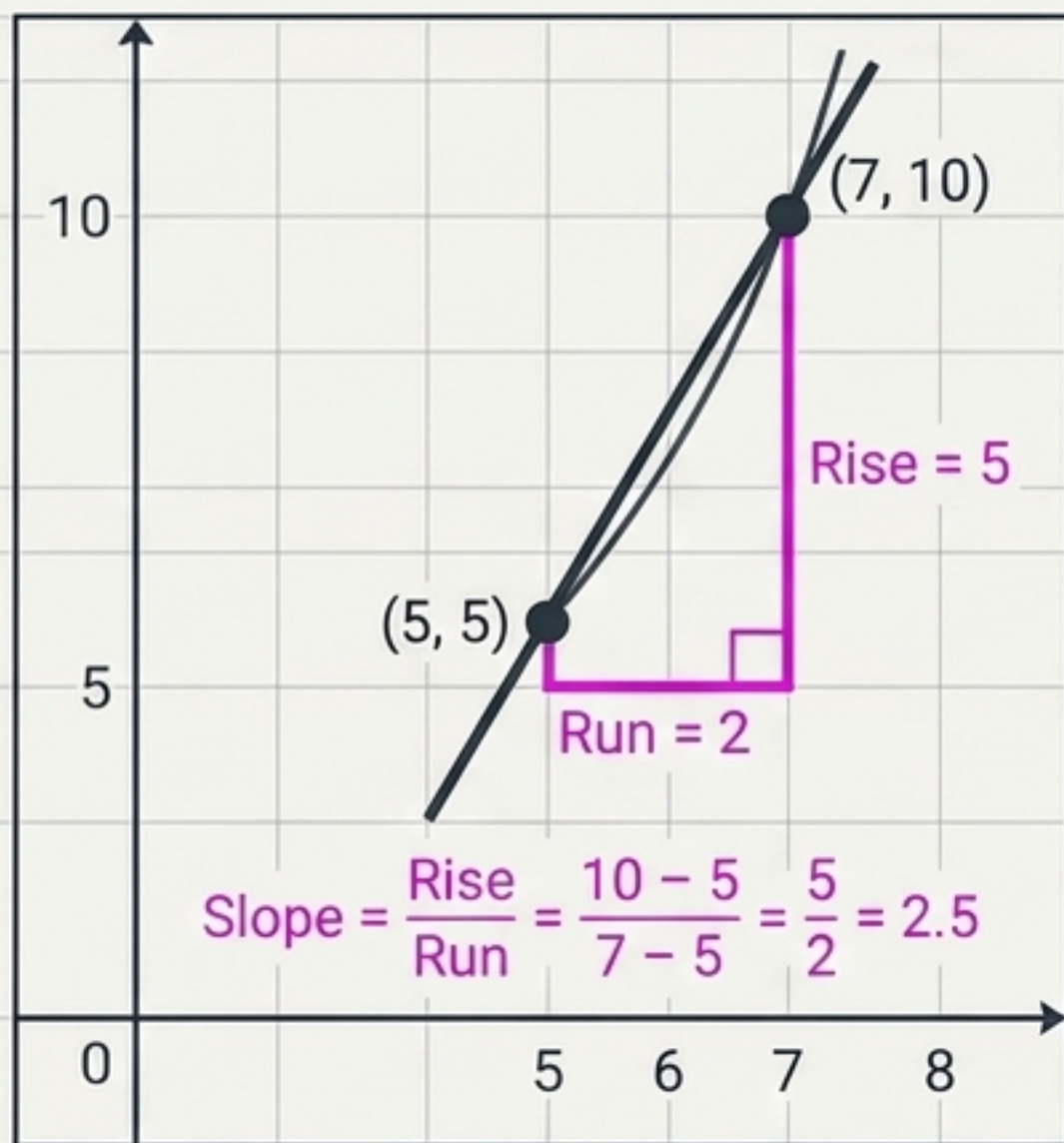
Geometric Intuition: The Secant Line

A line cutting through two distinct points on a curve calculates the average slope.

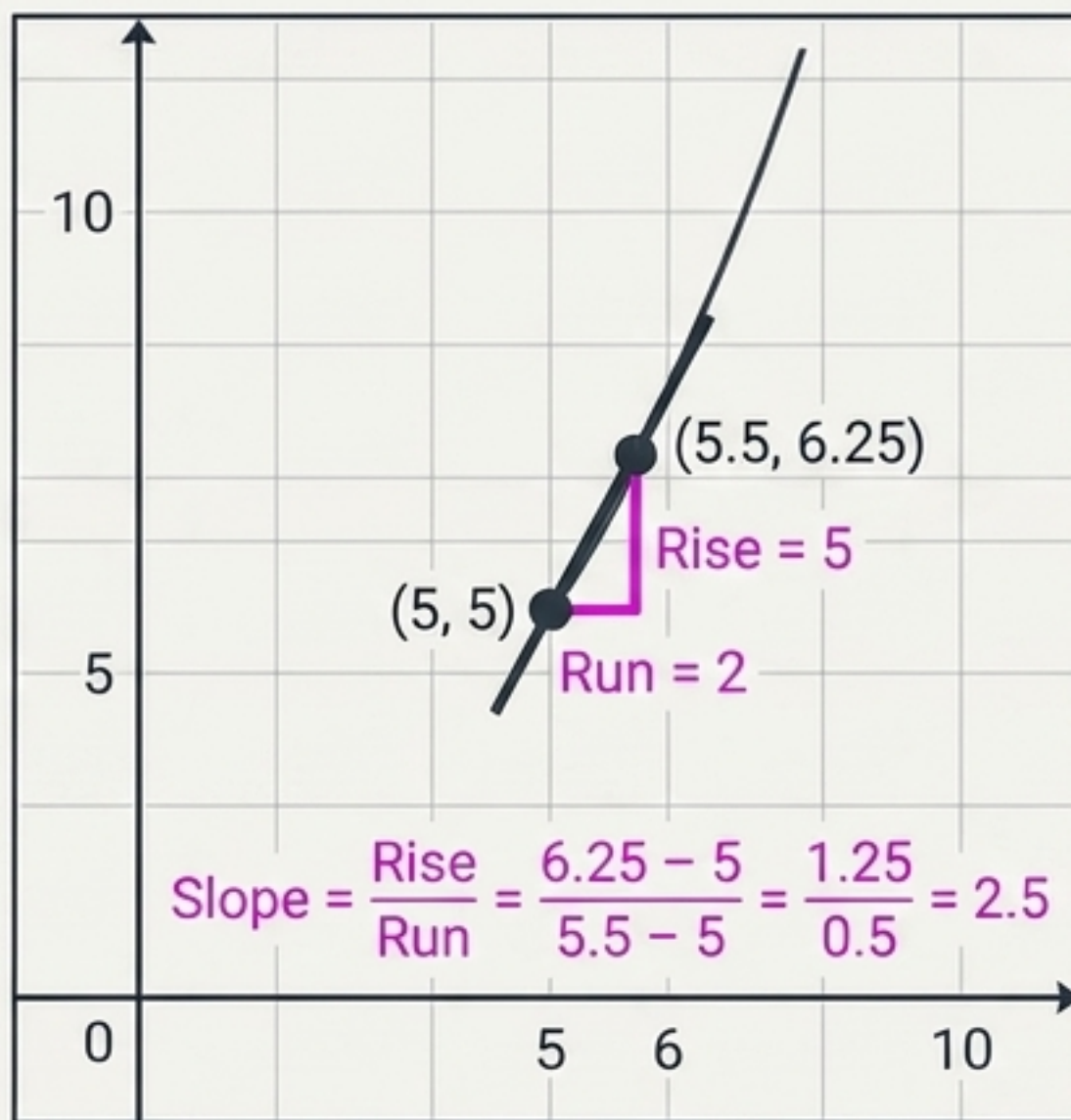


$$\begin{aligned}\text{Slope} &= \text{Rise} / \text{Run} \\ &= (10 - 5) / (7 - 5) \\ &= 5 / 2 = \mathbf{2.5}\end{aligned}$$

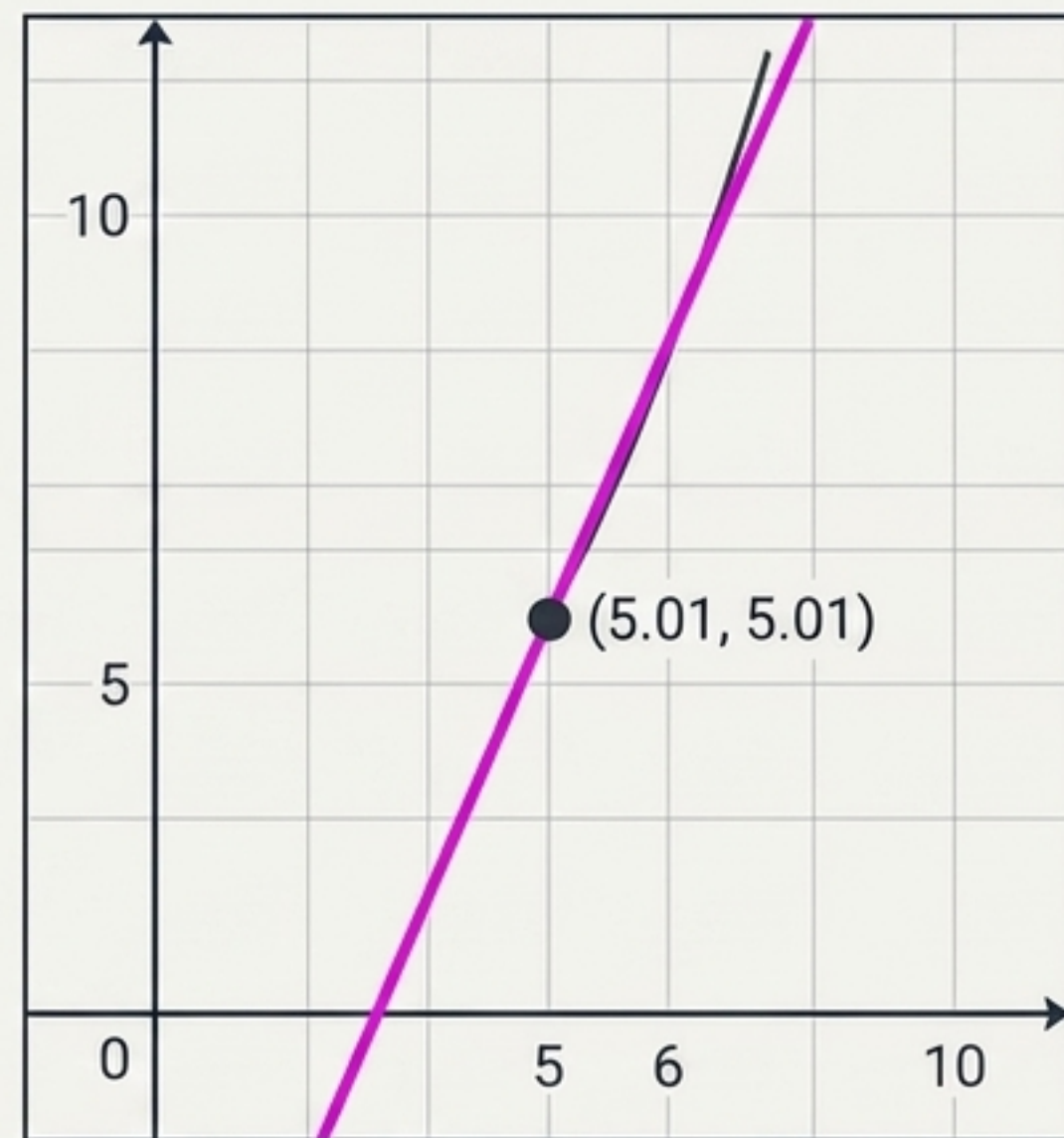
Shrinking the Triangle



1. Wide Interval



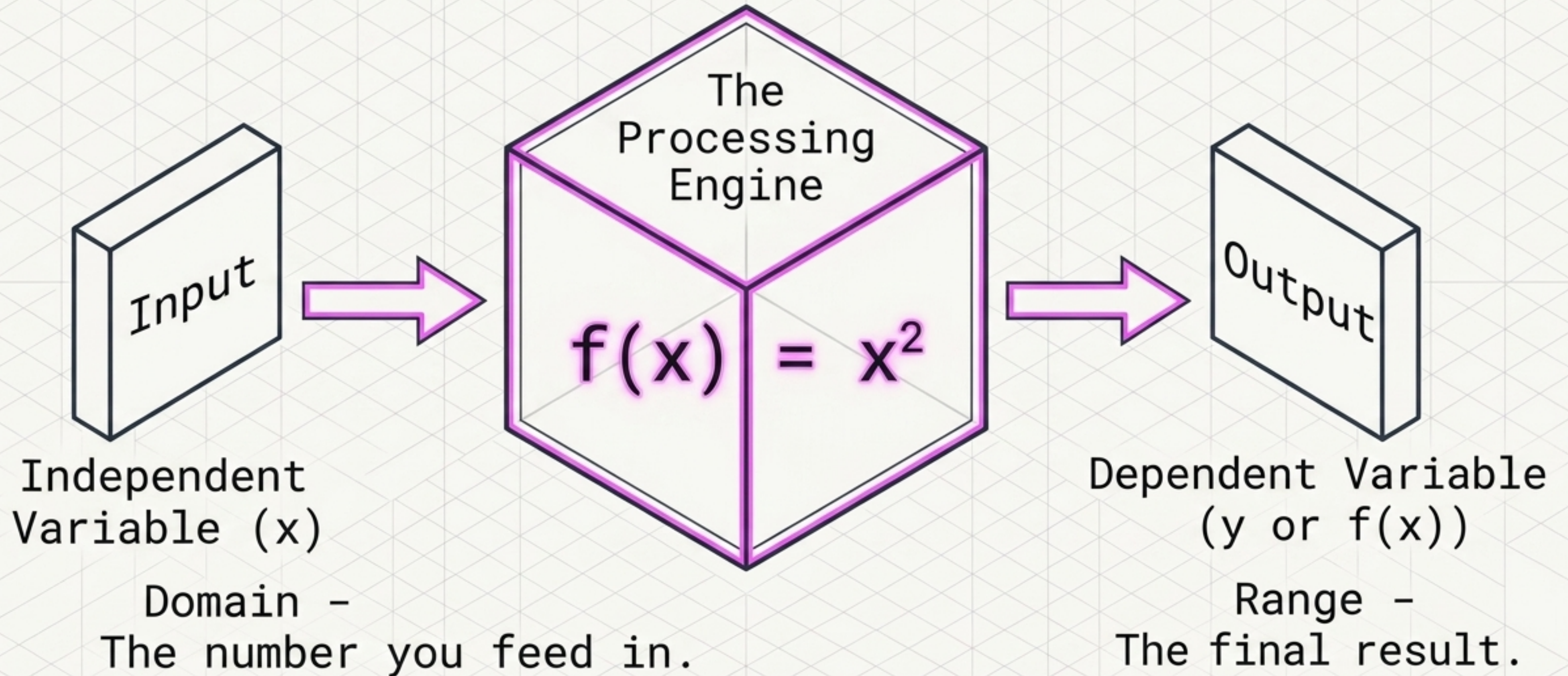
2. Shrinking Interval



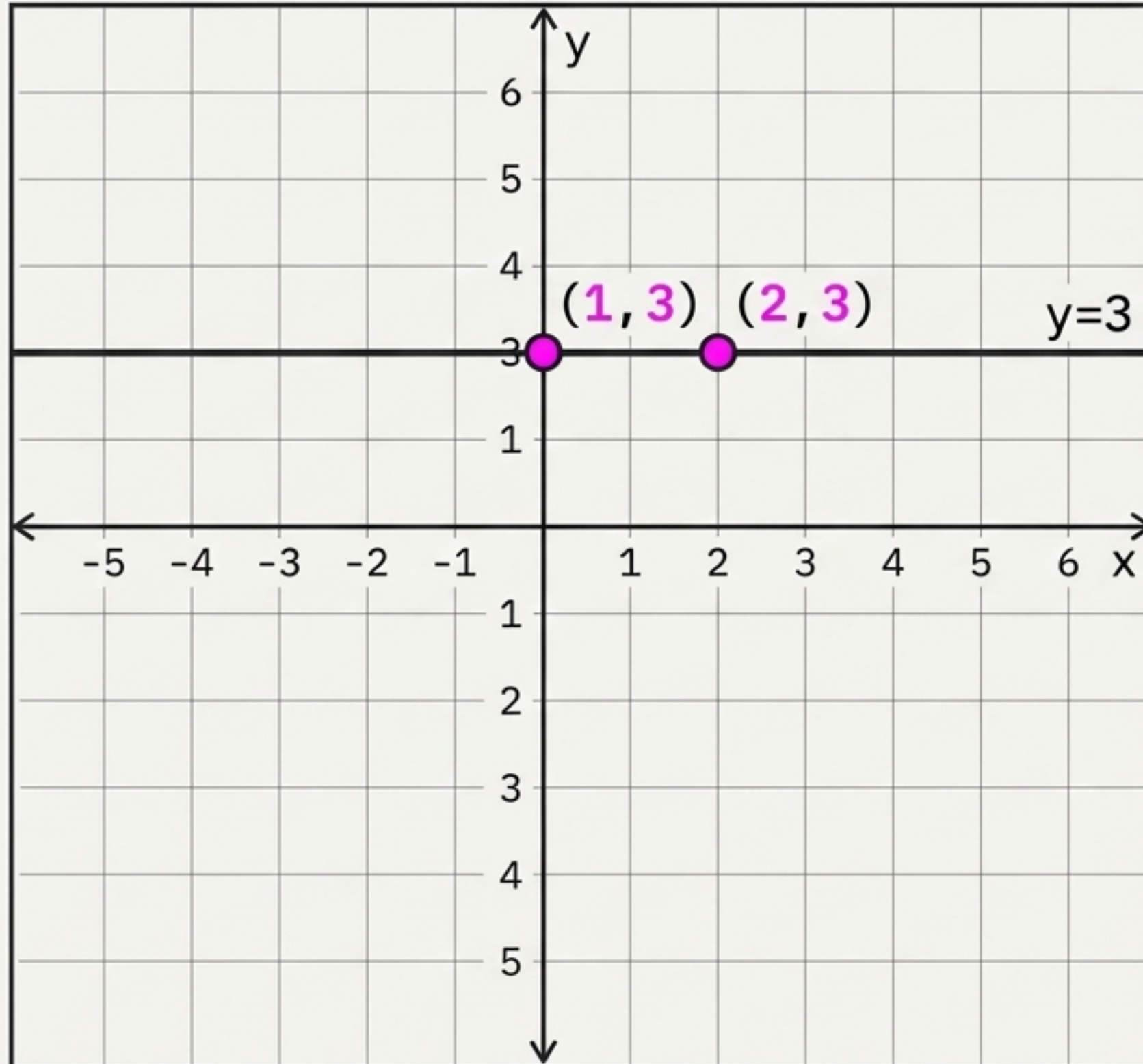
3. Infinitesimal Interval

When the **secant line** shrinks to a single point, it becomes the **Tangent Line**. Its slope is the **Derivative**!

Introduction to Functions: The Mathematical Machine



Rule 1: The Constant Function



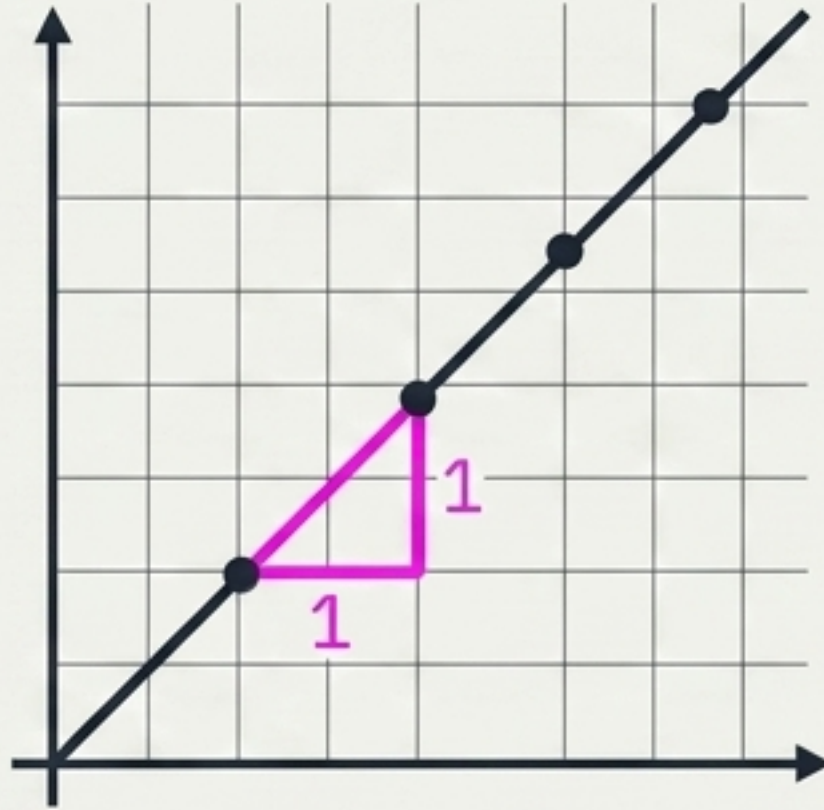
The Geometry: A horizontal line has absolutely no steepness.

Slope between any points (e.g., $x=1$ and $x=2$) is always $(3 - 3) / (2 - 1) = 0 / 1 = 0$.

The Universal Rule: The derivative of any constant c is always 0.

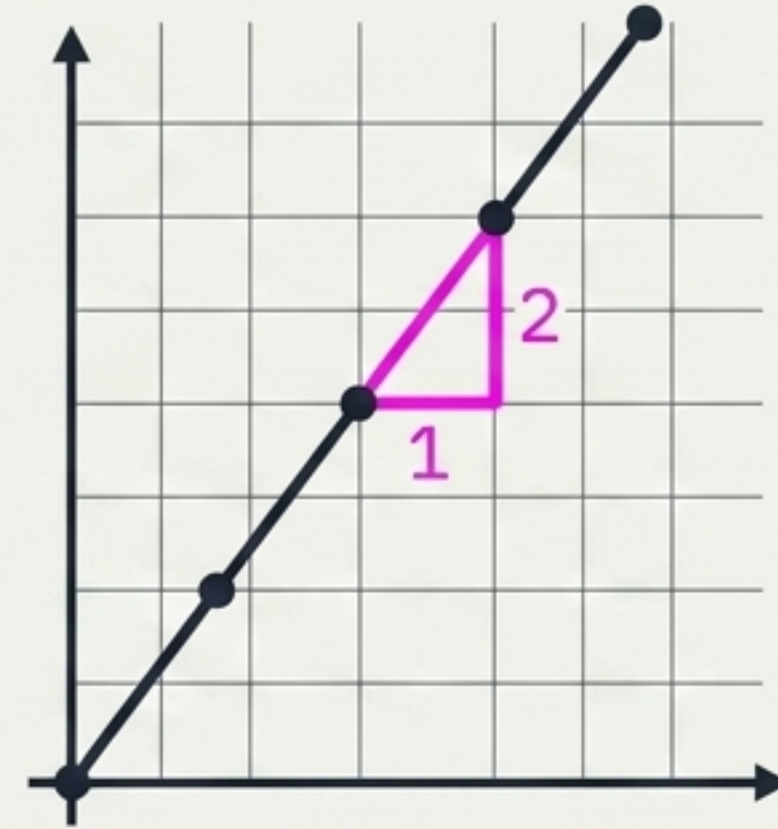
$$\frac{d}{dx}(c) = 0$$

Rule 2: Straight Lines



$$\text{Slope (m)} = 1$$

$$\frac{\Delta y}{\Delta x} = 1$$



$$\text{Slope (m)} = 2$$

$$\frac{\Delta y}{\Delta x} = 2$$

Key Insight: For any straight line, the instantaneous rate of change is always **constant** everywhere.

The Algebraic Proof (Using Limits)

Proving the derivative of $y = 2x$ is exactly 2 for any infinitesimally small step forward (Δx).

$$\frac{dy}{dx} = \frac{2(x + \Delta x) - 2x}{(x + \Delta x) - x}$$

$$\frac{dy}{dx} = \frac{2x + 2\Delta x - 2x}{\Delta x}$$

$$\frac{dy}{dx} = \frac{2\Delta x}{\Delta x}$$

$$\mathbf{dy/dx = 2}$$

The Universal Linear Rule

The derivative of any linear term mx is simply its coefficient, m .

$$\frac{d}{dx}(mx) = m$$

$$\frac{d}{dx}(5x) = 5$$

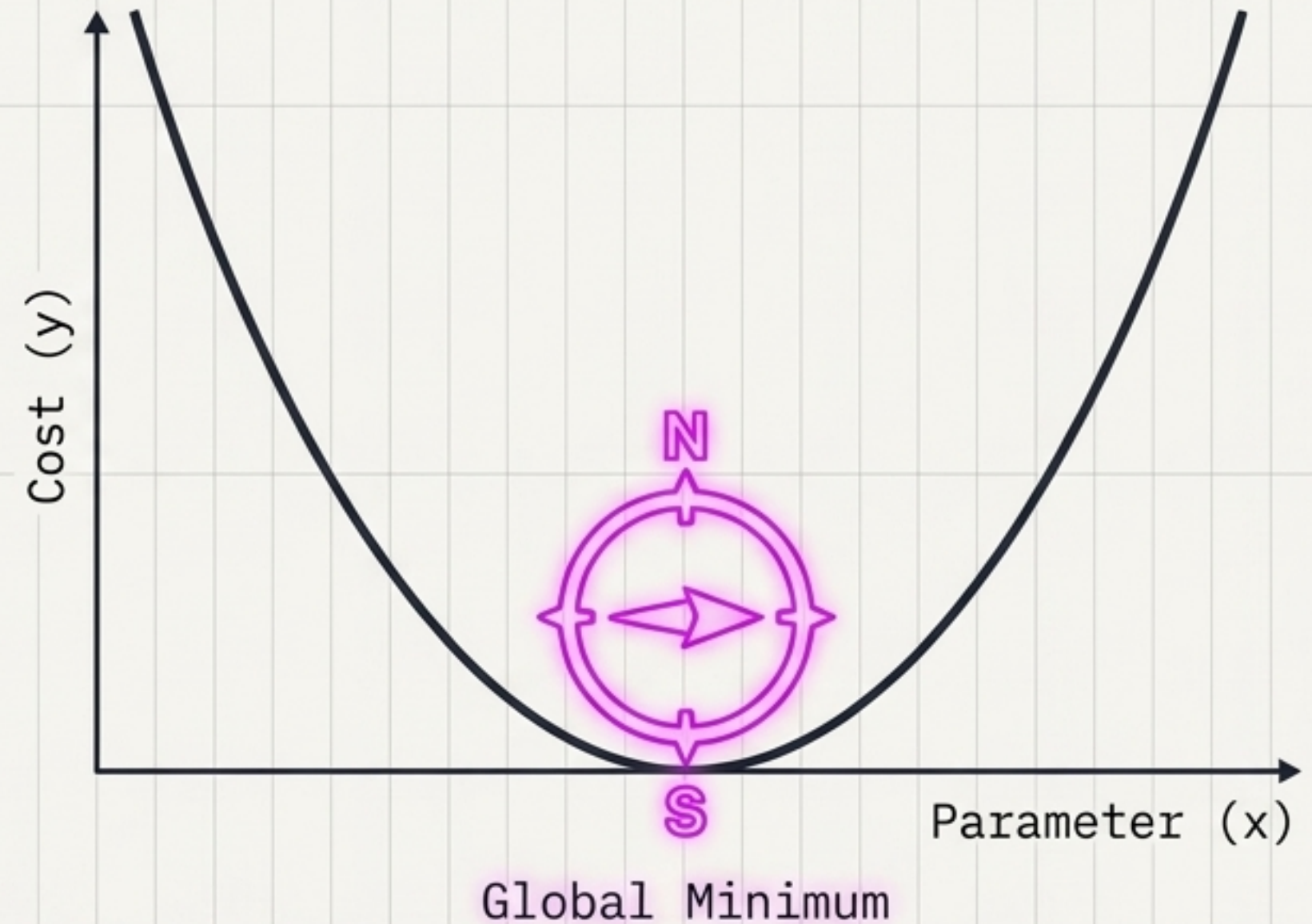
$$\frac{d}{dx}(-3x) = -3$$

$$\frac{d}{dx}(x) = 1$$

Tying It All Together

The Summary

1. **The Problem:** We measure Machine Learning mistakes using the Cost Function.
2. **The Guide:** We navigate toward zero mistakes using Derivatives (the instantaneous Tangent slope).
3. **The Engine:** We calculate those derivatives instantly using basic Algebraic Rules.



By mastering the calculus of change, you control the optimization engine of Machine Learning.